

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium(Assignment 1)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 40 (each student assigned distinct questions) Total Marks: 100

ANNEXURE A

Assignment 1

Code of Conduct

I (L. Sanjay Kumar/192425226) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L.S.Kumar

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding				
Algorithm				
Code Implementation				
Competitive Performance				
Test Case Handling				

Annexure A

Assignment 1

Question 28 – Social Media Hashtag Tracker

Problem Statement:

A social media company tracks trending hashtags.

Questions

- a) Develop a Java program using **HashMap**.
- b) Count hashtag frequency.
- c) Compare HashMap and TreeMap.
- d) Explain trend analysis.

(a) Java Program Using HashMap

```
import java.util.*;

public class HashtagTracker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        HashMap<String, Integer> map = new HashMap<>();

        System.out.print("Enter number of hashtags: ");
        int n = sc.nextInt();

        System.out.println("Enter hashtags:");
        for (int i = 0; i < n; i++) {
            String tag = sc.next();
            if (map.containsKey(tag))
                map.put(tag, map.get(tag) + 1);
            else
                map.put(tag, 1);
        }

        System.out.println("\nHashtag Frequency:");
        for (String tag : map.keySet())
            System.out.println(tag + " : " + map.get(tag));
    }
}
```

Sample Input

Enter number of hashtags: 6

Enter hashtags:

AI

Java

AI

ML

Java

AI

Sample Output

Hashtag Frequency:

AI: 3

Java: 2

ML: 1

(b) Count Hashtag Frequency

- Store each hashtag as the key in a HashMap.
- If the hashtag already exists, increment its count.
- Otherwise, insert it with a count of 1.
- The final values represent the frequency of each hashtag.

(c) Compare HashMap and TreeMap

HashMap	TreeMap
Stores data in random order	Stores data in sorted order
Faster ($O(1)$ average lookup)	Slower ($O(\log n)$ lookup)
Uses hash table	Uses Red-Black Tree
Allows one null key	Does not allow null keys
Best for fast searching	Best when sorted output is required

(d) Explain Trend Analysis

Trend analysis is the process of identifying the most frequently used hashtags over a period of time. By counting hashtag frequencies, social media companies can:

- Identify trending topics.
- Recommend popular content.
- Analyze user interests.
- Improve targeted advertisements.
- Monitor public opinion and events.